Sequences and Series

1. Recall that an *arithmetic sequence* is of the form

$$\begin{aligned}\{a_n\} &= \{a_1 + (n-1)d \mid n = 1, 2, 3, \dots\} \\ &= \{a_1, a_1 + d, a_1 + 2d, \dots\}\end{aligned}$$

where a_1 is the *initial term* and $d = a_k - a_{k-1}$ is the *difference*. **(a)** Show that the sum of the first k terms of an arithmetic sequence $\{a_n\}$ is given by

$$\sum_{i=1}^{k} a_i = \frac{k}{2}(a_1 + a_k).$$

(b) Based on (a), find the expression for the mean value of the arithmetic sequence $\{a_1, a_2, \dots, a_n\}$. **(c)** Based on (a), find the expression for the sum of the first n natural numbers $1 + 2 + 3 + \cdots + n$.

2. In general, a *power series* is of the form

$$a_0 + a_1 z + a_2 z^2 + \dots a_n z^n + \dots$$

with complex coefficients a_k and variable z. The simplest power series is the *geometric series*

$$1 + r + r^2 + \cdots + r^n + \dots$$

(a) Show that *partial sum* of the geometric series is given by

$$1 + r + r^2 + \dots r^{n-1} = \frac{1 - r^n}{1 - r}.$$

(b) Using your results from (a), show that an infinite geometric series converges if $|r| < 1$ and diverges if $|r| \geq 1$.

3. Notice that arithmetic sequences can be defined using a simple *recursion* formula

$$a_{n+1} = a_n + d,$$

where d is some constant. **(a)** Find the general recursive form for geometric sequences. **(b)** Define the *Fibonacci* sequence as

$$a_n = a_{n-1} + a_{n-2}.$$

Assuming that the limit

$$L = \lim_{n \to \infty} \frac{a_{n+1}}{a_n}$$

exists and is nonnegative, show that L must satisfy

$$L^2 - L - 1 = 0.$$

4. There are many ways to define *Euler's number* e, but one is via compound interest:

$$e = \lim_{n\to\infty} \left(1 + \frac{1}{n}\right)^n.$$

(Try to think about why it's called *compound interest.*) **(a)** Show that

$$e = \sum_{k=0}^{\infty} \frac{1}{k!}$$

where $k!$ denotes the *factorial* of k using the binomial theorem. **(b)** In general, the *exponential function*, denoted by $\exp z$ or e^z, can de defined as

$$e^z = \sum_{k=0}^{\infty} \frac{z^k}{k!}.$$

Once again verify this using the binomial theorem by evaluating

$$e^z = \lim_{n\to\infty} \left(1 + \frac{z}{n}\right)^n.$$

(c) *Euler's identity* states that

$$e^{i\theta} = \cos\theta + i\sin\theta$$

for any real θ. Using this identity, show that $\cos\theta$ and $\sin\theta$ have power series representations

$$\cos\theta = \sum_{n=0}^{\infty} \frac{(-1)^n\theta^{2n}}{(2n)!} = 1 - \frac{\theta^2}{2!} + \frac{\theta^4}{4!} - \dots$$

$$\sin\theta = \sum_{n=0}^{\infty} \frac{(-1)^n\theta^{2n+1}}{(2n+1)!} = \theta - \frac{\theta^3}{3!} + \frac{\theta^5}{5!} - \dots$$

(d) Show that

$$\tan\theta = -i\frac{e^{i\theta} - e^{-i\theta}}{e^{i\theta} + e^{-i\theta}}.$$

(e) Find $\lim_{n\to\infty} a_n$ and $\lim_{n\to\infty} b_n$ where

$$a_n = \operatorname{Re}\left(1 + \frac{i}{n}\right)^n, \quad b_n = \operatorname{Im}\left(1 + \frac{i}{n}\right)^n.$$